

BACH NGUYEN

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EDUCATION

Imperial Business School

London, UK | Aug 2024 - Nov 2025

MSc in Risk Management and Financial Engineering

Relevant modules: Financial Statistics, Stochastic Calculus for Finance, Financial Engineering, Empirical Finance, Risk Management and Valuation, International Finance, Computational Finance with C++, Applied Quantitative Macro Strategies, Systematic Trading Strategies with Machine Learning Algorithms

University of California, Irvine

Irvine, USA | Oct 2021 - Dec 2023

BA Business Economics (GPA: 3.8), Dean's List: 4 quarters

Relevant modules: Applied Econometrics I and II, Probability and Statistics, International Economics, Behavioral Economics

PROJECTS

SPX Volatility Risk Premium with HMM Regime Filtering | Python

London, UK | Apr 2026

- Built systematic iron condor strategy on SPX options (30 to 45 DTE, 16 delta short strikes, 50% credit close, 21 DTE roll, 1.5x stop loss) gated by a 5 state Gaussian HMM trained on macro features including SPY 21 day return, VIX, VIX3M to VIX term structure, HYG to LQD credit ratio and AAA to BAA spread
- Achieved walk forward out of sample Sharpe of 1.75 with 1.4% max drawdown across 120 cycles from 2015 to 2024, Deflated Sharpe passing at 420 trials, Newey-West HAC t statistic of 3.04 at lag 21 and Almgren-Chriss-Li capacity ceiling at GBP 500k to 1M; cross asset extension to DAX, TLT and GLD failed acceptance on correlation, IV proxy and scaling

Topological Graph Statistical Arbitrage with ML Signal Combination | Python

London, UK | Feb 2026

- Built a dollar neutral stat arb strategy on 135 US mid cap stocks by constructing KNN correlation graphs with Ledoit Wolf shrinkage, applying graph Laplacian smoothing to extract peer relative residual signals, and combining them with momentum, reversal and volatility features using LASSO and Random Forest for cross sectional rank prediction based on Gu, Kelly and Xiu
- Achieved 2.58% annualized alpha with near zero market beta (0.13) over 6.5 years out of sample, top bottom quintile spread of 11.3% annualized (t stat 1.55) and graph features like node degree and clustering coefficient making up 28% of total RF feature importance

BTC/ETH Statistical Arbitrage using Stochastic Processes and ML | Python

London, UK | Dec 2025

- Built a crypto pairs trading system using Kalman filter for dynamic hedge ratio, Ornstein-Uhlenbeck MLE for mean reversion calibration, and LSTM combined with XGBoost for signal prediction, with HMM regime filtering and momentum confirmation to avoid unfavorable market conditions
- Backtested on 12 months hourly data with Kelly criterion position sizing, reporting out of sample Sharpe of 0.28 after identifying and correcting look ahead bias in the target construction

PROFESSIONAL EXPERIENCE

APAC Pharmaceuticals

Ho Chi Minh city, Vietnam | Mar 2024 - Aug 2024

Finance Associate

- Oversaw preparation and analysis of financial statements to support strategic business decisions, identifying cost reduction opportunities that contributed to an average 10% decrease in operating expenses
- Negotiated bulk shipping rates and freight forward contracts, reducing logistics costs by 8%
- Managed import export related taxes and duties, ensuring compliance with local and international tax regulations and resolving previous compliance issues that had accounted for approximately 5% loss in revenue

Outlier

Remote, USA | Jan 2024 - Mar 2024

Economist

- Created and validated 200+ graduate level economics problems and solutions for training large language models, covering macroeconomics, econometrics and international trade
- Evaluated and ranked AI generated economic analyses for factual accuracy and logical consistency, providing feedback on reasoning errors that contributed to iterative model improvement

Maybank Kim Eng Investment Banking Group

Ho Chi Minh city, Vietnam | Mar 2022 - June 2022

Investment Banking Analyst Intern

- Supported the team in building financial models and projecting key metrics for client presentations, covering equity research across Vietnamese listed companies
- Assisted with risk assessment and scenario analysis on client portfolios and delivered daily market updates and stock analysis reports to subscribed clients in English

ADDITIONAL SKILLS

Technical Skills: Python (pandas, NumPy, SciPy, scikit learn, statsmodels, PyTorch, XGBoost, hmmlearn), C++, R, Git, EViews

Languages: Fluent: English, Vietnamese; Basic: Japanese

EXTRA-CURRICULAR

Competitions: IMC Prosperity 4 Global Trading Competition (Apr 2026), ranked 313 of 18,803 teams overall and 263 algorithmic;

Society: ICL Algorithmic Trading, Imperial Investment Society, Imperial Finance Society (ICFS); **Sports:** IC Boxing Club, Snowboarding, Golf